DATA VISUALISATION ASSIGNMENT – 2

STUDENT ID of Member 1 – 24241163
STUDENT ID of Member 2 – 20390573

Data collection process:

The data was collected by using the API key generated from Google Distance Matrix API. This collected data includes commute distance and time data from biggest towns in each county to Dublin city

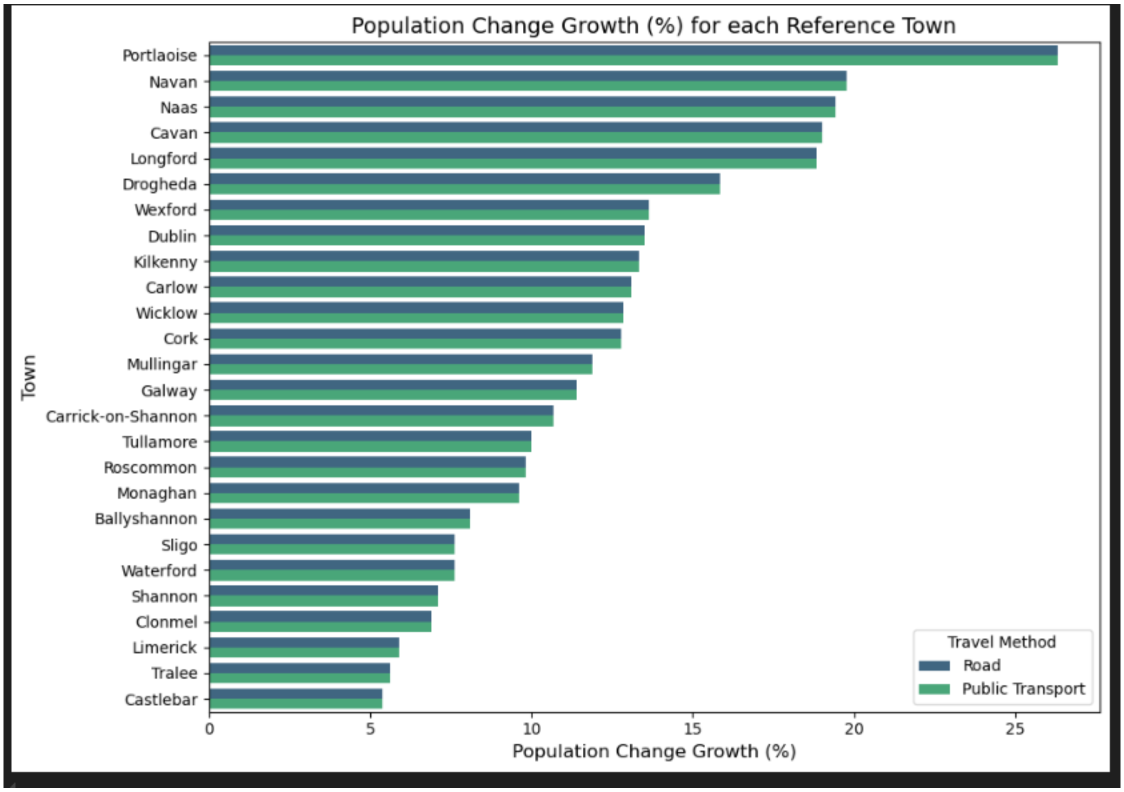
- **API key generation and configuration:-** An API key was generated from Google Cloud's Distance Matrix API. This API is essential for the travel data querying which allows the requests during the timestamp to be fetched.
- **Google Maps API and data Retrieval Process:-** The gmaps library in Python is a part of the Google Maps API client especially the Google Maps python client, which helps conveniently to interact with Google's mapping services which include the Google Distance Matrix API. This helps in retrieving the information from Google Maps.
- **Query Timestamps: -** Single timestamp has been for morning which will query at 9AM and in 7PM in the evening.
- **Methodology:-** Builds a query using the Google Distance Matrix API that has been collected. This API will return the Distance in meters and duration to commute in seconds. This data is formatted to display as Hours and Kilometres.
- **Consistency and Accuracy:-** Pre-determined data slots have been used here(9AM in the morning , 7PM in the evening)

The collected csv file consists of :

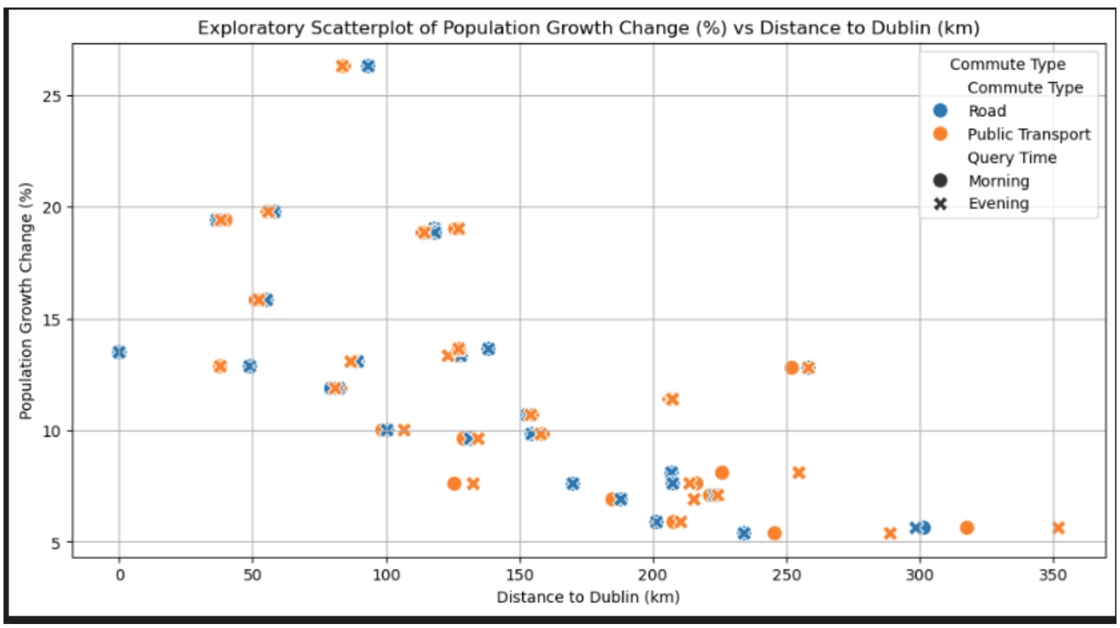
Columns	**Description**
County	**Displays the total 26 counties in Ireland**
Reference Town	**Takes the biggest town in each county**
Commute Type	**Driving or transit**
Distance to Dublin (km)	**Travel distance from each town to Dublin**
Commute Time (min)	**Time taken for travel**
Query Time	**When the query is sent Morning or evening**

The population growth data has been added to the csv file to do the Analysis.

Exploratory Graphs:

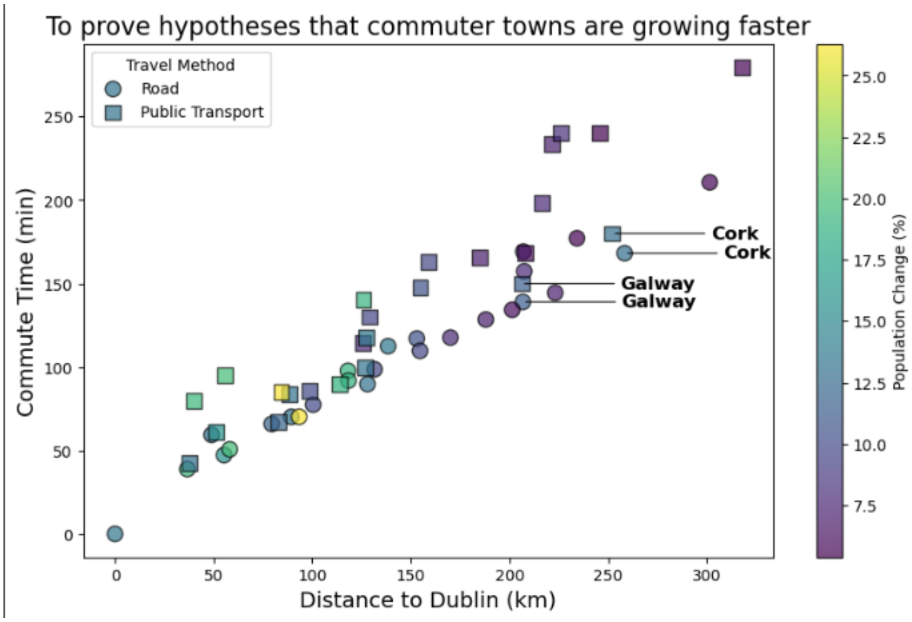


This chart shows the population growth of each of our reference towns. To anyone with knowledge of Irelands geography it seems to support our hypotheses that towns closer to Dublin had higher population growth. Two outliers we noticed were Cork and Galway and we assume that this is because both counties host cities of their own.



This graph suggests that the towns that are close to Dublin experience a higher population growth compared to the other towns. A relatively clear linear relationship can be seen here between population growth and distance from Dublin.

Explanatory Graphs:



It was fairly clear from the exploratory graphs created that the hypothesis that population growth increase is higher in commuter towns close to Dublin was correct. The bar graphs we created showed to anyone with some knowledge of geography in Ireland that counties in Leinster are growing faster. The graphs did show two outliers as well though. These outliers are Cork and Galway. However, the hypothesis still holds value. This is because both Cork and Galway contain large urban centres.
Due to these facts a scatterplot was chosen to show the relationship between distance, commute time and population growth. A scatterplot was chosen as it can show the linear relationship between commute time and distance and along with a colour scale can easily show population growth as well. Another pro for the scatterplot was that since the aim of the exercise was to show that distance and commute time from Dublin affect population growth, all of our data points did not need labels. We did apply labels to Galway and Cork datapoints to point out during the presentation that they are outliers and why.

Finally, the graph uses the viridis colour scale which is designed to be read by anyone including people with colour-blindness.

CONCLUSION:

This study shows that commute times and distances are typically shorter in areas with faster rates of population expansion. Because they are nearer to employment hubs and have superior transportation infrastructure, areas with greater population growth tend to have lower commute times. A self-sustaining cycle of expansion and shorter travel times is created when new residents are drawn in by the improved accessibility.

CONTRIBUTIONS:

24241163 – Data Collection, Created data collection part in Report, Created data collection part in PPT, Video for data collection part.

20390573 – Created graphs, wrote explanatory graph section in report, recorded video for exploratory and explanatory graphs.

Appendix:

Bar Chart 1:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

file_path = "Commute_Data_Ireland_final.csv"
Our_data = pd.read_csv(file_path)
data_sorted = our_data.sort_values(by="Population Change (%)", ascending=False)

plt.figure(figsize=(10, 7))
sns.barplot(palette="viridis", data=data_sorted, y="Reference Town", hue="Commute
Type", x="Population Change (%)")

plt.title("Population Change Growth (%) for each Reference Town", fontsize=14)
plt.ylabel("Town", fontsize=12)
plt.xlabel("Population Change Growth (%)", fontsize=12)
plt.legend(title="Travel Method", loc="lower right")

plt.tight_layout()
plt.show()
```

Explanatory Growth:

```
 data_filtered = data[data["Query Time"] != "Evening"]

data_filtered["Population Change (%)"] = ((data_filtered["Year_2016"] -
data_filtered["Year_2006"]) / data_filtered["Year_2006"]) * 100

plt.figure(figsize=(10, 6))

markers = {"Road": "o", "Public Transport": "s"}

for commute_type, marker in markers.items():
    subset = data_filtered[data_filtered["Commute Type"] == commute_type]
```

```python
    scatter = plt.scatter(
        subset["Distance to Dublin (km)"],
        subset["Commute Time (min)"],
        c=subset["Population Change (%)"],
        cmap="viridis",
        s=100,
        edgecolor='black',
        alpha=0.7,
        marker=marker,
        label=commute_type
    )

plt.colorbar(scatter, label="Population Change (%)")

plt.title("To prove hypotheses that commuter towns are growing faster",
fontsize=16)
plt.xlabel("Distance to Dublin (km)", fontsize=14)
plt.ylabel("Commute Time (min)", fontsize=14)

for _, row in data_filtered[data_filtered["Reference Town"].isin(["Galway",
"Cork"])].iterrows():
    x, y = row["Distance to Dublin (km)"], row["Commute Time (min)"]

    plt.annotate(
        row["Reference Town"],
        xy=(x, y),
        xytext=(x + 50, y),
        fontsize=12, fontweight="bold", color="black",
        ha="left", va="center",
        arrowprops=dict(arrowstyle="-", color="black", lw=0.8)
    )

plt.legend(title="Travel Method")

plt.show()
```